3.2 TEST OF SIGNIFICANCE FOR TWO NORMAL POPULATION VARIANCES

Test procedure:

This test compares the variances of two independent normal populations, *viz.*, $N(\mu_X, \sigma_X^2)$ and $N(\mu_Y, \sigma_Y^2)$.

Step 1 : **Null Hypothesis** $H_0 : \sigma_X^2 = \sigma_Y^2$

That is, there is no significant difference between the variances of the two normal populations.

The alternative hypothesis can be chosen suitably from any one of the following

(i) $H_1 : \sigma_X^2 < \sigma_Y^2$ (ii) $H_1 : \sigma_X^2 > \sigma_Y^2$ (iii) $H_1 : \sigma_X^2 \neq \sigma_Y^2$

Step 2 : **Data**

Let $X_1, X_2, \ldots, X_m$ and $Y_1, Y_2, \ldots, Y_n$ be two independent samples drawn from two normal populations respectively.

Step 3 : **Level of significance** α

Step 4 : **The test Statistic**

$F = \dfrac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} = \dfrac{S_1^2}{S_2^2}$ under H_0 and its sampling distribution under H_0 is $F_{(m-1,\, n-1)}$.

Step 5 : **Calculation of the Test Statistic**

The test statistic $F_0 = \dfrac{s_X^2}{s_Y^2}$

Step 6 : **Critical values**

H_1	$\sigma_X^2 < \sigma_Y^2$	$\sigma_X^2 > \sigma_Y^2$	$\sigma_X^2 \neq \sigma_Y^2$
Critical value(s) f_e	$f_{(m-1,\, n-1),1-\alpha}$	$f_{(m-1,\, n-1),\, \alpha}$	$f_{(m-1,\, n-1),1-\alpha/2}$ and $f_{(m-1,\, n-1),\, \alpha/2}$

Step 7 : **Decision**

H_1	$\sigma_X^2 < \sigma_Y^2$	$\sigma_X^2 > \sigma_Y^2$	$\sigma_X^2 \neq \sigma_Y^2$
Rejection Rule	$F_0 \leq f_{(m-1,\, n-1),\, 1-\alpha}$	$F_0 \geq f_{(m-1,\, n-1),\, \alpha}$	$F_0 \leq f_{(m-1,\, n-1),\, 1-\alpha/2}$ or $F_0 \geq f_{(m-1,\, n-1),\, \alpha/2}$

Note 1: Since $f_{(m-1,\, n-1),\, 1-\alpha}$ is not avilable in the given F-table, it is computed as the reciprocal of $f_{(n-1,\, m-1),\alpha}$.

i.e., $f_{(m-1,\, n-1),\, 1-\alpha} = \dfrac{1}{f_{(n-1,\, m-1),\, \alpha}}$

Note 2: A F-test is based on the ratio of variances, it is also known as Variance Ratio Test.

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